

Vowel Characters

Jojo has a string S and need you to tell how many vowel characters are there in the string. The list of vowel characters are "A", "I", "U", "E", "O" (both in its uppercase or lowercase form). Please note that "Y" is not considered as vowel character in this problem in any possible strings.

For this problem, please make a function "`int countVowel(const char *str)`" that return the number of vowel character in the given string.

Note that for this problem, you **HAVE TO** to use a **recursive solution**. **Do not** include "string.h" library or use any built-in C/C++ function. You need to make your own recursive function and you do not need any string.h related function in order to solve this problem.

Format Input

The input begins with an integer T , indicating the number of test cases. Each of the next T lines will contain a string S .

Format Output

For each test case, output "Case #X: Y". X is the number of test case starting from 1, and Y is the number of vowel character in the given string.

Constraints

$1 \leq T \leq 100$

$1 \leq |S| \leq 100$

$|S|$ is the length of string S

Sample Input	Sample Output
3 Halo there It's fun to use a function Goodluck	Case #1: 4 Case #2: 9 Case #3: 3

Note:

For some of you that questioned on why "Y" is explicitly stated as "not a vowel character", it's because of "Y" can be sometimes regarded as a vowel character in words like "fly", "my", "why", etc.. So, by stating the assumption upfront in this problem, those who know this fact can safely ignore the case of "Y" in "fly", "my", etc..